TEST PLAN SPECIFICATION

Document ID	OC-PLAN	Version	1.0
Project	OpenCart Platform	Status	Ready for Review
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1 Overview

This Test Plan defines the scope, objectives, approach, and deliverables for testing the OpenCart application to ensure critical business processes function as intended and meet defined quality expectations. Testing will focus on validating core application functionality, including authentication, product browsing, cart operations, checkout, and order processing, through a structured, risk-based approach. Test activities will support defect detection, coverage assessment, and quality evaluation to provide confidence in system stability and release readiness.

2 Scope

Category	Details
In Scope	**Core User Management** • User registration (Sign Up) • User authentication (Sign In / Sign Out) • Password recovery and reset functionality • Session and access control validation **Catalog and Navigation** • Homepage and global navigation components (header, footer, menus) • Category listing pages and product detail pages (PDP) • Product search, sorting, filtering, and pagination **Shopping Features** • Wishlist functionality • Shopping cart operations (add, update, remove items) **Checkout and Order Processing** • Checkout workflows • Shipping, payment, and order confirmation flows **API and Data Validation** • API request/response validation • Database verification for critical business transactions and data integrity **Automation Testing** • Automation of critical user journeys • Regression test coverage for high-risk functionality **Performance Testing** • Performance validation for key user transactions and critical system behavior under expected load

Out of Scope	• Administration and back-office functionality testing • Third-party integrations beyond functional checkout validation, including external payment gateway provider systems • Deep security assessments, including penetration testing and vulnerability exploitation testing • Native mobile application testing is excluded; testing covers the web application only, including supported desktop and mobile browser access.

3 Entry & Exit Criteria

Criteria	Details
Entry Criteria	• Approved requirements and user stories are baselined and available. • Test environment is configured, stable, and accessible. • Test data and required test accounts are prepared. • Test cases are designed, reviewed, and approved. • Application build is deployed and ready for test execution. • Required test tools and automation framework are available and operational. • API endpoints, supporting interfaces, and required database access are available.
Exit Criteria	• At least 95% of planned test cases have been executed. • All critical and high-priority test cases have passed. • No open critical or high-severity defects remain. • Remaining defects are documented, accepted, and have agreed mitigation plans. • Regression testing (including automated) has been completed successfully. • Test summary report completed and stakeholder approval obtained.

4 Test Environment

Component	Details
Operating Systems	Windows 10 and Windows 11 (latest stable updates) Android (latest stable via Chrome mobile browser)
Browsers	Google Chrome (latest stable version) Microsoft Edge (latest stable version)
Network Conditions	Stable internet connection (Wi-Fi)

5 Testing Approach

This section defines the overall testing approach for the OpenCart e-commerce application. The objective is to ensure the system meets its functional and non-functional requirements across all critical business flows, including user management, product catalog, shopping cart, and checkout processes.

Testing will be performed using a combination of static, manual, automated, and API-level validation across supported environments to ensure comprehensive coverage, early defect detection, and product quality assurance.

Testing Type	Activities
Static Testing	• Review of requirements, user stories, and acceptance criteria • Validation of completeness, consistency, and testability of specifications • UI/UX design review for consistency and usability alignment • Early identification of ambiguities, gaps, and logical defects
Manual Testing	• User authentication (registration, login, logout, password reset) • Product catalog (browsing, search, sorting, filtering, pagination) • Shopping cart (add, update, remove items) and Wishlist • Checkout process (shipping, payment, order confirmation) • UI validation (layout, alignment, messages, navigation behavior)
API Testing	• REST API endpoint validation (request/response correctness) • Authentication and authorization verification • Data integrity and response validation • HTTP status codes validation (positive and negative scenarios) • Error handling and boundary conditions
Automation Testing	• Automation of critical end-to-end user journeys (login, cart, checkout) • Smoke and regression test suite execution • Data-driven testing for multiple input combinations • Continuous execution support for stable builds • Maintainable and reusable test scripts
Database Testing	• Validation of CRUD operations • Verification of data consistency across related tables • Validation of order, cart, and user-related data integrity • Verification of constraints and data rules • Ensuring no data corruption during transactions
Non-Functional Testing	Compatibility Testing • Cross-browser testing (Google Chrome, Microsoft Edge) • Web access validation on Android via Chrome browser • Layout and functional consistency across supported environments Usability & UI Testing • Navigation flow consistency • UI elements validation (labels, buttons, messages) • Basic accessibility checks (forms, tab navigation, readability) Regression Testing • Re-execution of core test suite after changes or fixes • Ensuring existing functionality remains unaffected

Testing Type	Activities
	Smoke & Sanity Testing • Smoke testing for build stability verification • Sanity testing for targeted feature validation after fixes

6 Test Levels

- Component Testing
- Integration Testing
- System Testing
- Acceptance Testing

7 Test Deliverables

Category	Deliverables
Test Documentation	• Test Plan Document • Test Scenarios Document • Manual Test Cases Document • Requirement Traceability Matrix (RTM)
Test Execution Artifacts	• Test execution results (manual and automated) • Test run reports from test management tools (e.g., Jira / Xray / Zephyr) • Test execution summary reports
Defect Management	• Defect/Bug reports (maintained in Jira and/or Excel as backup) • Defect lifecycle tracking reports • Defect analysis and severity distribution reports
Automation Deliverables	• Automation test framework (Selenium-based) • Automated test scripts (UI regression suite) • Automation execution reports
API Testing Deliverables	• API test cases • Postman collections / REST Assured test suites • API execution reports and defect validation results
Database Deliverables	• Database validation scripts and SQL queries • Data integrity verification reports • Database test execution results
Final Reporting & Closure	• Test Summary Report • Final test documentation package

8 Test Phases

Phase	Activity	Duration	Responsible
1	Planning & Preparation Test	1 DAYS	Testing Lead
2	Test Analysis & Design	3 DAYS	Testing Team
3	Test Environment Setup	1 DAYS	Testing Team
4	Test Execution & Defect Management	5 DAYS	Testing Team
5	Test Closure & Reporting	2 DAYS	Testing Lead

9 Team Members and Roles

Team Structure

The QA team consists of five members working collaboratively on all testing activities for the OpenCart project. All members participate equally in Manual, Automation, API, and Database Testing activities.

Name	Role	Responsibilities
Mohamed Ahmed	QA Team Member	Participates in manual testing, automation execution, API validation, database testing, defect reporting, and regression cycles.
Mohamed Ashrafa	QA Team Member	Participates in manual testing, automation execution, API validation, database testing, defect reporting, and regression cycles.
Mariam Elgandour	QA Team Member	Participates in manual testing, automation execution, API validation, database testing, defect reporting, and regression cycles.
Maryam Saleh	QA Team Member	Participates in manual testing, automation execution, API validation, database testing, defect reporting, and regression cycles.

Shorouk Mohamed	QA Team Member	Participates in manual testing, automation execution, API validation, database testing, defect reporting, and regression cycles.

10 Risks & Mitigation Plan

Risk	Impact	Mitigation Strategy
Test environment instability or downtime	High	Maintain a backup test environment and coordinate early environment readiness checks before execution.
Cross-browser compatibility issues	Medium	Execute testing across multiple browsers (Chrome, Edge) and validate critical UI flows on each supported configuration.
Limited testing time due to tight deadlines	Medium	Prioritize high-risk and business-critical test cases using risk-based testing approach and execute in prioritized cycles.
Flaky automation test results	Low	Regularly review and maintain automation scripts, and isolate unstable test cases from regression suite until stabilized.

11 Test Metrics & KPIs

Category	Metric	Description	Target
Test Execution	Test Case Execution Progress	Percentage of executed test cases out of total planned	≥ 95%
	Test Case Pass Rate	Percentage of passed test cases from executed tests	≥ 90%
	Test Case Fail Rate	Percentage of failed test cases	≤ 10%
	Blocked Test Cases	Percentage of test cases blocked due to issues	≤ 5%
Defect Metrics	Critical Defect Leakage	Number of critical defects escaped to production	0
	Defect Reopen Rate	Percentage of defects reopened after being marked fixed	≤ 10%
	Defect Rejection Rate	Percentage of rejected defects (invalid,	≤ 5%

Category	Metric	Description	Target
		duplicate, not a bug)	
	Severity Distribution	Distribution of defects by severity before release	No open Critical or High
Test Coverage	Requirements Coverage	Percentage of requirements covered by test cases	100%
	Test Case Coverage	Coverage of application features by test cases	≥ 95%
	Risk Coverage	Coverage of high-risk areas identified in risk analysis	All high-risk areas
Automation Metrics	Automation Coverage	Percentage of critical test scenarios automated	≥ 60%
	Automation Pass Rate	Percentage of passed automated test cases	≥ 90%
	Flaky Tests Rate	Percentage of unstable or inconsistent automated tests	≤ 5%
	Regression Execution Time Reduction	Improvement in regression execution time over cycles	Continuous improvement
Performance Metrics	Response Time	Average response time for key transactions	≤ 3 seconds
	Error Rate Under Load	Percentage of errors under expected load conditions	≤ 2%
	System Stability	System behavior under expected load conditions	No critical failures
Quality KPIs	Release Readiness Index	Overall readiness level based on quality metrics	High / Medium / Low
	Entry Criteria Compliance	Degree of meeting entry criteria before test execution	100%
	Exit Criteria Compliance	Degree of meeting exit criteria before release sign-off	100%

12 References

- OpenCart Demo Website: https://awesomeqa.com/
- **User Stories**